# What Crosses Borders?

# The New York Cash Session and the Cross-Section of Asia-Pacific Transmission

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## Abstract

A companion paper (Kinoshita, 2026) shows, within a "t-0.5" framework, that New York close information predicts the next-day direction of Asia-Pacific markets and that this predictability concentrates in the opening gap. This paper asks two sharper questions: which part of the New York trading day carries the information, and where in the cross-section of markets and sectors it lands. Using walk-forward logistic regression on daily data (December 2009 to January 2024) for nine Asia-Pacific equity markets and the seventeen TOPIX-17 Japanese sector ETFs — predicting each target's next-day opening-gap direction from New York close information — we document three results. First, decomposing the New York day into its overnight gap and its cash session, the cash-session (open-to-close) return alone reproduces 98.6% of the full close-to-close predictive power, whereas the overnight gap captures far less; the cash session dominates in all 26 targets (intraday minus overnight = +6.6 percentage points on average). A proof-of-concept using S&P 500 E-mini futures shows that the post-close, pre-Tokyo futures drift adds essentially nothing beyond the cash close, ruling out live-futures leakage. Second, in the cross-section, directional predictability rises with a sector's sensitivity to global equity risk: opening-gap accuracy = 60.1 + 16.3×β ($R^2 = 0.49$, $p = 0.002$). Third, broad risk factors absorb sector-specific information for sixteen of seventeen sectors; only energy exhibits a distinct sector-specific carrier (Japanese energy paired with the U.S. energy ETF: +6.6 points, about 6σ). What crosses borders is specifically the New York cash session, and it lands in proportion to global-risk exposure, with energy as a commodity-driven exception.

**Keywords:** international information transmission; price discovery; directional accuracy; overnight gap; sector cross-section; index futures; market efficiency

**JEL:** G14, G15, F36, C53

# 1. Introduction

When New York closes, its final prices are widely believed to set the tone for the next trading day in Asia. A companion paper (Kinoshita, 2026) makes this precise. Within a "t-0.5" timing framework — in which t = 0 is the New York close (06:00 Japan Standard Time, JST), t = 0.5 is the Tokyo open (09:00 JST), and t = 1 is the next New York close — it shows that New York close information (the S&P 500, the VIX, the ten-year Treasury yield, and the dollar-yen rate) predicts the next-day direction of Asia-Pacific markets out of sample, and that this predictability is concentrated in the opening gap rather than in the intraday session. The natural next questions are not whether transmission occurs, but its anatomy: which part of the New York trading day actually carries the predictive information, and where, across markets and sectors, that information comes to rest.

This paper answers both. We keep the empirical engine of the companion paper — a walk-forward logistic regression predicting the sign of each target's next-day opening gap from New York close information — and apply two decompositions. The first is on the source side: we split the New York day into its overnight gap (previous close to open) and its cash session (open to close), and ask which component drives Asia's open. The second is on the destination side: we widen the set of targets from four to nine Asia-Pacific markets and, more importantly, to the seventeen Japanese TOPIX-17 sector portfolios, and ask which are predictable and why.

Three findings emerge. First, the carrier is the New York cash session. Across all twenty-six targets, the cash-session return alone reproduces 98.6% of the full close-to-close predictive accuracy, while the overnight gap alone captures markedly less; the cash session beats the overnight component in every single target, by 6.6 percentage points on average. A proof-of-concept using continuously traded S&P 500 E-mini futures shows that the futures drift after the New York cash close and before the Tokyo open adds nothing beyond the cash close, which rules out the alternative that our predictor is merely proxying near-contemporaneous live futures. Second, the cross-section of predictability is orderly: cyclical, export-oriented sectors are more predictable than domestic, defensive ones, and a single variable — a sector's beta to the S&P 500 — explains about half of the variation (gap accuracy = $60.1 + 16.3\times\beta$, $R^2 = 0.49$). Third, broad risk factors already absorb sector-specific information almost everywhere; the lone exception is energy, for which a sector-specific U.S. energy signal adds large, statistically singular predictive power.

Taken together, these results refine the companion paper's headline. What crosses borders is not New York close information in the abstract, but specifically the move of the U.S. cash session, which lands on the Asia-Pacific open in proportion to each market's or sector's exposure to global equity risk, with energy standing apart as a commodity-driven exception. The remainder of the paper develops these results in turn.

## 2. Related literature and framework

The transmission of returns and volatility across international equity markets is a long-standing subject. Early work established contemporaneous and lagged linkages among national indices (Eun and Shim, 1989; Hamao, Masulis, and Ng, 1990; King and Wadhwani, 1990), and a substantial volatility-spillover literature followed, typically using GARCH-family or vector-autoregressive models (Lin, Engle, and Ito, 1994; Karolyi, 1995). That literature largely asks how much variance is transmitted, and treats correlation or volatility, not the out-of-sample predictability of direction, as the object of interest.

The companion paper reframes the question as one of directional predictability measured out of sample, and introduces the t-0.5 timing convention that lets New York close information be evaluated as a genuine ex-ante predictor of the Tokyo open. It also connects to the machine-learning-in-finance literature (Gu, Kelly, and Xiu, 2020; Fischer and Krauss, 2018), while showing that a parsimonious, interpretable logistic model is difficult to beat here. The present paper builds directly on that framework and contributes three things the prior literature does not provide: a decomposition of the predictor's own trading day that isolates the cash session as the carrier; a systematic cross-section of transmission across markets and sectors; and a simple asset-pricing regularity — the beta law — that organizes that cross-section.

## 3. Data and method

All data are daily and come from Yahoo Finance via the yfinance interface, spanning 1 December 2009 to 1 January 2024. Predictors are the New York close values of four series — the S&P 500 (SPX), the VIX, the ten-year Treasury yield (TNX), and the dollar-yen exchange rate (USDJPY) — entered as daily log returns. Following the t-0.5 convention, each is observable at the New York close (06:00 JST) and hence before the Tokyo open (09:00 JST). Because the predictor set is finalized at 06:00 JST while every target opens no earlier than roughly 08:00-10:30 JST, there is a strict one-directional gap between predictor and outcome: look-ahead or target leakage is structurally impossible, not merely controlled for.

For every target we form the next-day opening gap, gap = log(open / previous close), and take its sign as the outcome. Models are estimated by walk-forward logistic regression: at each date the classifier is refit on a rolling window of past observations (2,800 days in the market analysis; 1,250 days in the sector and decomposition analyses, to preserve out-of-sample length for shorter series), features are standardized using only the training window to avoid look-ahead, and the one-step-ahead sign is predicted. We report directional accuracy (DA), the out-of-sample share of correct sign predictions, together with a z-test of DA against the random-walk benchmark of 0.50 and, for sectors, 95% Wilson confidence intervals. Because thinly traded instruments occasionally carry a

stale opening price equal to the previous close, we flag the share of such days and, where it is high, restrict interpretation to the close-to-close measure. In every specification, days on which the opening price exactly equals the previous close — a zero opening gap — are dropped from the target before estimation, so the reported accuracies are not mechanically inflated by stale quotes.

Two decompositions structure the analysis. On the source side, the New York day for the S&P 500 is split into its overnight component, ON = log(open / previous close), its cash session, ID = log(close / open), and their sum, the close-to-close return FULL = ON + ID. On the destination side, targets comprise nine Asia-Pacific markets (Japan, Korea, Hong Kong, Australia, India, Taiwan, Singapore, Indonesia, Malaysia) and the seventeen TOPIX-17 Japanese sector ETFs. Finally, to probe the three hours between the New York cash close and the Tokyo open, we use hourly S&P 500 E-mini futures (ES=F) over the freely available recent window as a proof of concept. Free intraday futures data are available only for a recent two-year window, which is why the test in Section 5 is a proof of concept rather than a full-sample result.

## 4. Which part of the New York day crosses? The cash-session result

We first decompose the predictor's own trading day. For each of the twenty-six targets we predict the next-day opening gap using, in turn, the New York overnight component alone (ON), the cash session alone (ID), the full close-to-close return (FULL), and the two components jointly (BOTH). If Asia's open reprices information that is already set when New York opens, ON should dominate; if it reprices the U.S. cash-session move, ID should dominate. Throughout this section the predictor is the S&P 500 alone, split into its components; the four-variable model used elsewhere (Section 6, Table 2) is a separate specification, so the accuracies here and there are not directly comparable.

**Table 1. New York day decomposition: out-of-sample directional accuracy (%) predicting the next-day opening gap. ON = overnight (prev. close→open), ID = cash session (open→close), FULL = close-to-close. *ASX opening gap unreliable (high stale-open share).**

| Target | ON | ID | FULL | ID−ON |
|---|---|---|---|---|
| KOSPI (KR) | 62.2 | 75.8 | 75.7 | +13.6 |
| NIFTY 50 (IN) | 68.2 | 75.5 | 74.9 | +7.3 |
| Nikkei 225 (JP) | 61.4 | 71.9 | 74.1 | +10.5 |
| TAIEX (TW) | 64.0 | 71.2 | 73.1 | +7.3 |
| STI (SG) | 58.1 | 70.6 | 71.5 | +12.5 |
| Jakarta (ID) | 65.2 | 70.5 | 71.2 | +5.3 |

| | | | | |
|---|---|---|---|---|
| Hang Seng (HK) | 59.1 | 69.9 | 69.6 | +10.8 |
| ASX 200 (AU)* | 57.6 | 67.0 | 66.5 | +9.4 |
| KLCI (MY) | 57.2 | 64.1 | 65.0 | +7.0 |
| Sector mean (17) | 60.1 | 65.4 | 66.5 | +5.3 |
| Summary — ID > ON | | | | 26 / 26 |

The pattern is unambiguous and universal. The cash session (ID) beats the overnight component (ON) in every one of the twenty-six targets, by 6.6 percentage points on average. More strikingly, the cash session alone reproduces 98.6% of the accuracy of the full close-to-close return, and adding the overnight component on top of it raises accuracy by less than one point on average (+0.97). The overnight gap is therefore largely redundant once the cash session is known: the information Asia prices at its open is the U.S. cash-session move, not the state of the world when New York opened.

The economics are coherent. The overnight-into-open component reflects Asian and European trading that precedes the U.S. session; by the time Tokyo opens the following morning, that information is both partly its own echo and stale relative to the full U.S. day that has since elapsed. The cash session, by contrast, is the genuinely new U.S. information and is the closest in time to the Tokyo open. The one systematic nuance is geographic: markets exhibit stronger cash-session dominance than sectors, and India — the most time-distant market — shows an unusually informative overnight component, consistent with its "overnight" window mechanically containing more of the New York day. Put differently, what Tokyo prices at its open is not New York's opening level — which still embeds the echo of Tokyo's and Europe's own prior moves — but the U.S. cash-session move, a genuinely domestic American price discovery that Asia has not yet seen.

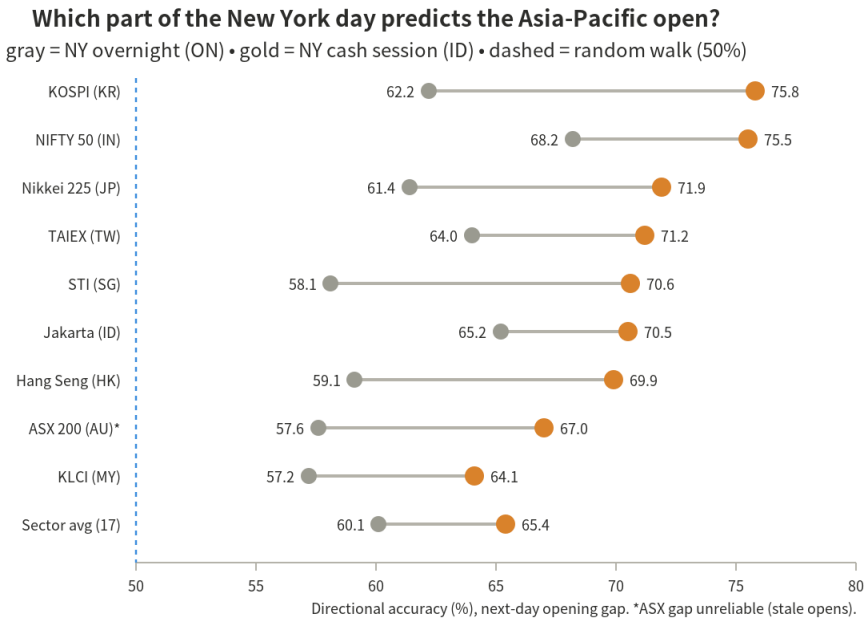



**Figure 1. The New York day decomposition. Each row connects a target's overnight accuracy (ON, gray) to its cash-session accuracy (ID, gold); the cash session is higher in every one of the 26 targets, and reproduces 98.6% of the full close-to-close accuracy on average.**

## 5. Ruling out live-futures leakage: the overnight futures session

A natural objection to any New-York-close predictor is that S&P 500 futures trade almost continuously, so a close-based signal might merely proxy a live futures price that keeps moving into the Tokyo open. The companion paper addresses the mirror-image concern by controlling for Nikkei futures; here we control on the U.S. side. Using hourly E-mini futures over the recent two-year window as a proof of concept, we construct a futures analogue of the New York close (the futures level around 06:00 JST) and the post-close drift up to one hour before the Tokyo open (06:00 to 08:00 JST).

The futures analogue of the close achieves DA = 72.6% ($p < 0.001$), closely reproducing the cash-based accuracy and validating the pipeline. The post-close drift, however, adds nothing: combining it with the close-based signal changes accuracy by −0.4 points (72.2% versus 72.6%). The drift on its own reaches only 59.3%, and even that figure is inflated by endogeneity, since the 08:00 JST futures level sits just one hour before the open and partly co-moves with it. The predictive information is thus locked in at the New York cash close; the subsequent futures session, the American analogue of the Osaka night session, carries no additional signal for the Tokyo open. This rules out live-futures leakage and anchors the predictability in genuine close-time information.

We stress that this test uses only a short, recent sample; a full-history intraday-futures replication is left for future work.

## 6. The cross-section of transmission: markets and sectors

We next widen the destination. Extending the four markets of the companion paper to nine, New York close information predicts the next-day direction of every Asia-Pacific market considered at the 0.1% level, and in each case the opening-gap accuracy exceeds the close-to-close accuracy — the gap-concentration result replicated across nine independent markets. Absolute accuracy varies with distance and market structure but never approaches the random-walk benchmark.

**Table 2. Japanese TOPIX-17 sectors: opening-gap directional accuracy (%) with 95% Wilson confidence intervals, and each sector's beta to the S&P 500, ranked by accuracy. All sectors significant at $p < 0.001$.**

| Sector | Gap DA | 95% CI | β |
|---|---|---|---|
| Electric & precision machinery | 72.0 | [69.8, 74.1] | 0.54 |
| IT & services | 70.5 | [68.2, 72.7] | 0.48 |
| Automobiles & transport equip. | 70.1 | [67.9, 72.2] | 0.49 |
| Machinery | 69.9 | [67.6, 72.1] | 0.61 |
| Materials & chemicals | 69.5 | [67.3, 71.7] | 0.50 |
| Banks | 68.9 | [66.7, 71.1] | 0.43 |
| Steel & nonferrous | 68.9 | [66.6, 71.0] | 0.58 |
| Trading & wholesale | 68.5 | [66.3, 70.7] | 0.61 |
| Construction & materials | 68.5 | [66.2, 70.6] | 0.45 |
| Real estate | 67.2 | [64.9, 69.4] | 0.54 |
| Pharmaceuticals | 66.4 | [64.1, 68.6] | 0.31 |
| Financials (ex-banks) | 66.3 | [63.9, 68.5] | 0.53 |
| Transportation & logistics | 66.2 | [63.8, 68.5] | 0.34 |
| Retail | 66.1 | [63.6, 68.5] | 0.41 |
| Energy resources | 64.1 | [61.8, 66.4] | 0.50 |
| Foods | 63.4 | [60.9, 65.8] | 0.29 |

| | | | |
|---|---|---|---|
| Electric power & gas | 61.6 | [59.2, 64.0] | 0.19 |

The sector cross-section is orderly. All seventeen sectors are individually significant, but they arrange themselves along an economic gradient: cyclical, export-oriented sectors — electric and precision machinery, machinery, automobiles, materials, steel — sit at the top, while domestic, defensive sectors — electric power and gas, foods, construction — sit at the bottom, roughly ten points lower. The confidence intervals overlap within the top cluster, so the fine ranking is not identified; what is identified, and robust, is the cyclical-versus-defensive gap.

## 7. Explaining the cross-section: the beta law

The economic gradient suggests a single organizing variable: a sector's sensitivity to global equity risk. We measure this as the sector's beta to the S&P 500 and regress opening-gap accuracy on beta across the seventeen sectors. The relationship is strong and statistically clear: Beta here is a lagged reaction beta — the slope from regressing a sector's daily return on the previous day's S&P 500 return — which is the transmission quantity of interest rather than a contemporaneous CAPM beta. We label this the Overnight Information Beta, denoted beta^OI, defined as the slope in r(i,t) = a_i + beta^OI_i * r(SPX, t-1) + e(i,t), where r(i,t) is sector i's daily close-to-close return and r(SPX, t-1) the previous New York close-to-close return; it is distinct from a contemporaneous CAPM beta.

$$\textit{Gap DA (\%)} = 60.07 + 16.26 \times \beta, \quad r = 0.70, \quad R^2 = 0.49, \quad p = 0.0018.$$

Beta alone accounts for roughly half of the cross-sectional variation in directional predictability. The slope implies that a sector with beta 0.6 is about seven points more predictable at the open than one with beta 0.2, and the intercept near 60 indicates that even a near-zero-beta sector retains meaningful predictability, reflecting a common component that reaches every sector. Economically, sectors more exposed to global risk import more of the U.S. cash-session move into their opening price, which is exactly the quantity that transmits. The in-sample relationship is not a small-sample fluke: a 5,000-shuffle permutation test returns p = 0.001 and a bootstrap 95% confidence interval for the slope of [6.9, 22.1], which excludes zero. Under a stricter disjoint-sample design - beta estimated on the first half of the sample and accuracy measured out of sample on the second, so that no mechanical overlap remains - the slope stays positive (+6.7) but loses significance (R^2 = 0.13, permutation p = 0.15), an attenuation that mixes the removal of any mechanical correlation with the well-documented instability of market betas across regimes. To separate genuine transmission from both, we replace the estimated beta with an exogenous, real-economy proxy for global exposure. For the manufacturing sectors we take overseas sales and production ratios from the METI and JBIC surveys of Japanese firms' overseas activities (for example, autos about 48%,

electrical and electronics about 41%, chemicals about 31%, foods about 23%); the non-manufacturing sectors, which those surveys do not cover, are assigned exposure levels from their industry structure. The relationship survives: opening-gap accuracy rises with this exposure measure (R^2 = 0.26, permutation p = 0.04, bootstrap 95% confidence interval for the slope [2.2, 15.9], excluding zero), and among the five sectors whose exposures are fully survey-based the correlation is strong (r = 0.80), though that subset alone (n = 5) is too small for significance. Because the manufacturing exposures come from external, non-market data, this addresses the tautology concern directly; but the modest overall fit and the estimated exposures for non-manufacturing sectors lead us to read it as supportive rather than decisive. We therefore treat the beta law as a descriptive, well-supported cross-sectional regularity rather than a proven causal relationship, with a definitive test awaiting firm-level foreign-revenue data across all sectors.

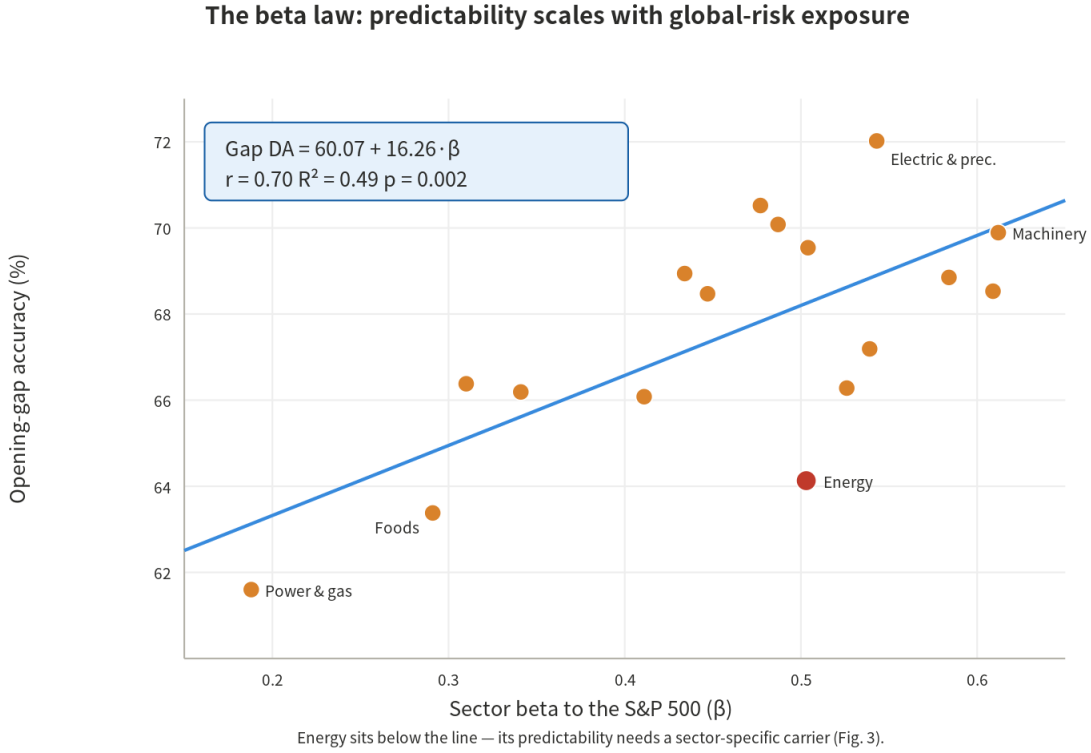



**Figure 2. The beta law. Across the 17 sectors, opening-gap accuracy rises with the sector's beta to the S&P 500 (Gap DA = 60.07 + 16.26·β; $R^2$ = 0.49, p = 0.002). Energy (red) lies below the line — the residual that the sector-specific analysis in Section 8 explains.**

## 8. The energy exception: sector-specific carriers

If broad factors like the S&P 500 and the VIX are the carrier, does any sector require its own, sector-specific U.S. signal? To answer this we add each of the nine long-history U.S. Select Sector ETFs, one at a time, to the baseline four-factor model for each Japanese sector, and record the change in opening-gap accuracy. Across the resulting matrix of 153 additions, the mean effect is zero (mean +0.01 point, standard deviation 1.06) and no sector-specific ETF helps its matched Japanese sector by more than about two points — with one exception.

That exception is energy. Adding the U.S. energy ETF to the Japanese energy sector raises opening-gap accuracy by about 6.6 points, roughly 6.7 standard deviations above the matrix mean and several times the next-largest effect. The U.S. energy ETF helps only energy: its effect on the other sixteen sectors averages about +0.08 point and never exceeds roughly 1.4. In both directions of the matrix - down the energy column and across the energy row - the energy-energy cell is a singular peak. For every other sector, the broad risk factors have already absorbed whatever the sector-specific U.S. signal could add. This exception dovetails with the beta law: energy is the sector whose predictability the beta regression most under-predicts, and the sector-specific U.S. energy signal is precisely what fills that gap. Energy transmits through a commodity-specific channel that the broad equity factors do not span.

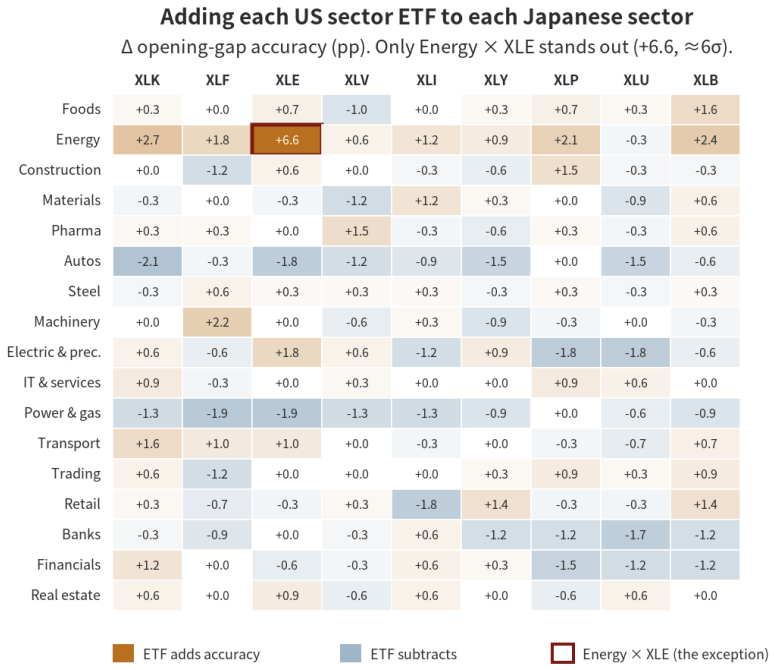



**Figure 3. Sector-specific carriers. Each cell is the change in a Japanese sector's opening-gap accuracy when the corresponding U.S. sector ETF is added to the baseline model. Only Energy × XLE (+6.6 points, boxed) stands out; every other cell lies within noise.**

A further question is not only whether energy carries a distinct signal but when, within the New York day, that signal arrives. We repeat the source-side decomposition of Section 4 on the energy-specific channel itself: rather than splitting the S&P 500 into its overnight and cash-session components, we split the U.S. energy ETF (XLE) the same way and add each component, one at a time, to the four-variable baseline model for Japan's energy resources sector. The baseline model alone attains DA = 64.1%, matching Table 2. Adding XLE's overnight gap (previous close to open) raises accuracy to 66.0% (+1.9 points); adding XLE's cash-session return (open to close) instead raises it to 68.9% (+4.8 points); and adding the full close-to-close return raises it to 70.9% (+6.7 points), matching the effect reported above. As a placebo, the same three additions to the machinery sector's baseline model (69.9%, matching Table 2) move accuracy by less than half a point in either direction (−0.4 to +0.3), confirming that the boost is specific to energy and not an artifact of adding any U.S. sector series.

The energy-specific channel therefore shows the same qualitative pattern documented for the broad market in Section 4 — the cash session contributes more than the overnight gap (+4.8 versus +1.9 points) — but the split is less one-sided: roughly a quarter of the full effect is already present at the New York open, consistent with crude-oil-relevant news reaching markets before the U.S. cash session begins (for example, during European trading hours), with the remainder accumulating during the session itself, plausibly including the Wednesday-morning EIA petroleum inventory report released about an hour after the New York open. This timing decomposition refines rather than overturns the energy exception documented above, and is reported here as a robustness check on, and elaboration of, that result.

## 9. Discussion

The three results compose a single sentence. What crosses borders is the move of the New York cash session; it lands on the Asia-Pacific open in proportion to each market's or sector's exposure to global equity risk; and it does so almost entirely through broad risk factors, with energy the one commodity-driven exception that requires its own carrier. The overnight gap is largely an echo, the post-close futures session adds nothing, and the intraday session of the destination market is close to efficient — so the entire transmission is compressed into the repricing that occurs at the open. Mechanically, this points to the U.S. cash session as the locus of genuine price discovery: it is during New York's liquid trading hours that globally relevant assets are repriced, and it is that repricing, not the thinner overnight tape, that Asia imports at its open.

These findings sharpen rather than overturn the efficient-markets reading of the companion paper. The predictability is directional, out of sample, and concentrated at a single instant — the open — which is where a market that has been closed overnight would be expected to incorporate accumulated foreign information. That the effect is anchored at the cash close, survives a futures control, and scales with global-risk exposure is consistent with rational repricing rather than a persistent, exploitable anomaly, particularly once transaction costs and the thin liquidity of the pre-open window are taken into account. The accuracy levels — for example 75.8% for KOSPI and 71.9% for Nikkei — are high relative to the 50–55% typically reported for out-of-sample daily close-to-close direction in the machine-learning literature, but the comparison is not like-for-like: the high figures reflect the choice of target (the low-noise opening gap), which is the paper's point, rather than any superiority of the model.

## 10. Limitations

Several limitations bound the results. The futures test is a proof of concept on a short, recent sample; a definitive statement requires full-history intraday futures data, which are not freely available. The sector analysis rests on ETF data whose out-of-sample windows are shorter than the market series, and some sectors carry non-trivial stale-open shares, so the fine sector ranking is only suggestive. The source-side decomposition uses the S&P 500 alone, since the opening prices of the VIX, Treasury, and currency series are less well defined; extending the decomposition to those series is left for future work. The sector-sweep result should be read with multiple-comparison caution, although the energy effect is far outside the range attributable to chance. Finally, directional accuracy is not a trading result: it does not net transaction costs, slippage, or the limited liquidity of the pre-open window, and the paper's purpose is to measure information transmission, not to propose a strategy. In particular, the companion paper shows that the VIX, the Treasury yield, and the dollar-yen rate each carry independent predictive content; extracting the intraday component of those series — whose opening prices are harder to define than equities — could raise accuracy further and is left for future work. The beta law is a cross-sectional regression on seventeen non-independent sectors with an estimated regressor, so its $R^2$ should be read as descriptive; we therefore assess it with permutation tests, bootstrap confidence intervals, and a disjoint-sample design (beta estimated on an earlier period, accuracy measured out of sample on a later one) that breaks the mechanical link between the two quantities. Relatedly, in the disjoint-sample design a lower R-squared need not imply that the original relationship was spurious: sector betas are themselves time-varying across regimes (for example the 2022 rate-hike period), so attenuation there conflates the removal of mechanical correlation with genuine beta instability. The bootstrap we report resamples the cross-section of seventeen sectors; a time-series block bootstrap that re-estimates both quantities would be needed to fully propagate the generated-regressor error

and is left for future work. The exogenous-exposure check in Section 7 uses published overseas ratios for the manufacturing sectors but estimated exposures for the non-manufacturing ones; assembling a uniform, firm-level foreign-revenue dataset across all sectors is the natural next step to move that result from suggestive to definitive.

## 11. Conclusion

Do bulls and bears move across borders? They do — but the object that crosses is specific. It is the move of the New York cash session, not the state of the world when New York opened, and not the futures drift that follows the close. It lands on the Asia-Pacific open in proportion to global-risk exposure, captured to first order by a single beta, and it travels through broad risk factors everywhere except energy, which carries its own commodity-specific signal. By decomposing both the source and the destination of overnight transmission, the paper turns a qualitative claim about spillovers into a quantitative anatomy of what crosses, when, and where it lands.

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